What survives diffusion noise

The EM chapters: the results

Brief for the meeting with Jerome and Marc · 23 September 2026

Every number below is a macro read from a frozen run; the derivations and definitions are in the compendium, the audit trail in the audit report.

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1 The question and the laboratory

A diffusion model learns the score of a noised data law, $\nabla_x \log P_t(x)$, and for real data nobody knows the target. We built a family where it is known exactly and is not trivial: a first-order Markov chain of length L with non-Gaussian innovations (Laplace, Student-t, uniform, bimodal, a one-parameter family in between), noised coordinate by coordinate. Because the noise attaches one factor per site, the posterior stays a chain and two sweeps of exact message passing return the score. The covariance of every innovation family is the same by construction, so the laboratory separates second-order structure (correlation ρ) from higher-order structure (the shape of the innovation law) cleanly. *Compendium, Chapters 4, 7, 11.*

Three questions were asked of it. What information survives the noise, order by order? Can the dynamics be learned from noised sequences alone, and how efficiently? And what does an estimator that knows the structure buy against a neural denoiser that does not, and where does that stop?

2 What stands, in one line each

1. **Noise is a low-pass filter on statistical order.** The Hermite moment of degree m is attenuated exactly by e^{-mt}; the Fisher information about a feature first visible at degree m decays as e^{-2mt}; the number of sequences needed for a fixed local precision grows as e^{2mt}. Correlation lives at degree two, shape at degree four. Verified on the exact score, with the constants. *Compendium, Chapters 3, 5, 10.*

2. **Second-order closure has an exact price.** Gaussian message closure computes the LMMSE of the covariance twin (Proposition 1), and the gap to the true score is $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$ to leading order. Adding the fourth-order term back closes 0.961 of the gap at $t = 0.68$ and all of it from $t = 2$. *Compendium, Chapters 10, 13.*

3. **The dynamics are learnable through the noise, and the information curve says where learning stops.** From 512 sequences the shape is recovered at the reference up to $t = 0.4$; fits split at $t = 0.7$ (4 of 16 collapse to a Gaussian innovation) and collapse at $t = 1$, one level below the point $t = 0.98$ where the Cramér–Rao reference crosses the coordinate's range. The correlation survives to the crossover. *Compendium, Chapters 5, 15.*

4. **Capacity buys exactly the fourth-order structure.** A Gaussian-innovation kernel lands on the LMMSE floor (0.167 against 0.168); an eight-component mixture reaches 0.023 at $M = 2048$. Across the generalised-Gaussian family the gain is ordered by κ_4 and reverses at the Gaussian member. The contrast holds in all 15 (ρ, L) cells of a regime sweep at 16 seeds, the paired difference running from -0.253 to -0.047 with every interval excluding zero, and in every cell the $C = 1$ arm lands on the LMMSE floor to 0.0099, which checks the whole apparatus. *Compendium, Chapters 13, 15, 17.*

5. **Against networks, the advantage is largest at equal information.** Under the standard recipe (clean sequences, fresh corruption every step) the best of six architectures reaches 0.060 at $M = 2048$, a paired factor 2.83 behind EM–BP. At EM–BP's own interface (one noised view per sequence, no clean chain, no true score for model selection) the best network reaches 0.165,

a paired factor 7.9 [6.6, 9.3]; given the clean sequence as a target but still one corruption, 0.155, a factor 7.3. *Compendium, Chapters 8, 17, 19.*

6. **That comparison now holds across regimes, and it is not about heavy tails.** Repeated at 6 combinations of correlation and chain length, with the networks re-screened inside each regime and 16 seeds per cell, EM–BP is ahead in every cell and in every seed, by paired factors from 4.6 to 10.3. Repeated at the frozen point with a platykurtic innovation instead of a heavy-tailed one, the factor is 8.8 [7.4, 10.2], the same as on the Laplace chain. What the structure buys is fourth-order structure of either sign. *Compendium, Chapters 17, 19.*

7. **The advantage ends where the structure is wrong, at a strength we can compute.** Under an AR(2) coefficient or a site-dependent ρ the ordering reverses inside the swept range; a rank-one global latent or a long-range coupling never reverses it. The large-sample error of the misspecified EM–BP is a closed-form Kullback–Leibler projection bias $b(\varepsilon)$ that reproduces the measured errors to within 0.019. *Compendium, Chapter 18.*

3 What survives the noise, order by order

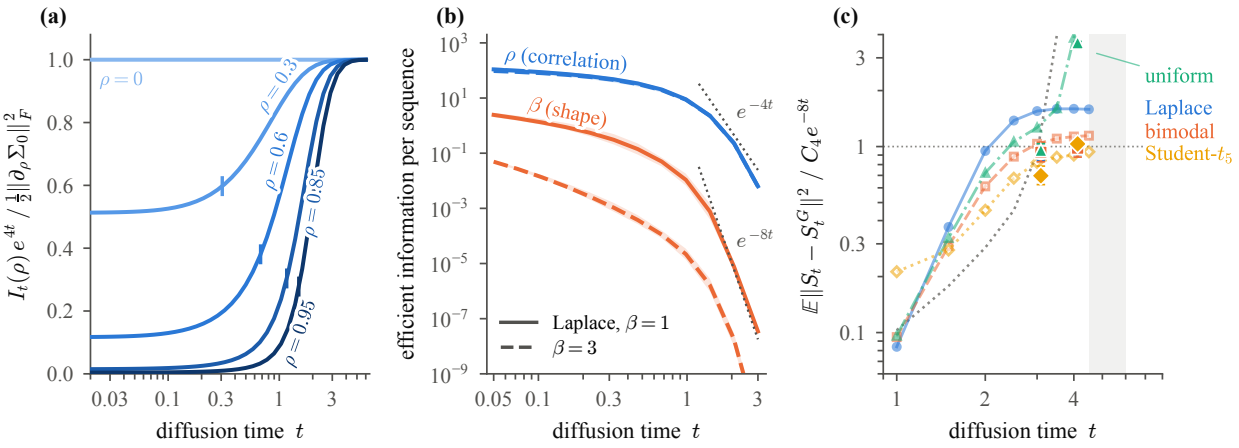


Figure 1: *(a)* Exact Fisher information about the correlation in the Gaussian chain, divided by its predicted asymptote $\frac{1}{2}\|\partial_\rho \Sigma_0\|_F^2 e^{-4t}$: every curve reaches 1, and the crossover $t_* = \frac{1}{2}\log \lambda_{\max}(\Sigma_0)$ (ticks) says when. *(b)* Information per sequence about the correlation (degree two) and the innovation shape (degree four) of the generalised-Gaussian chain, against the guides e^{-4t} and e^{-8t}. *(c)* The price of second-order closure, $\mathbb{E}\|S_t - S_t^G\|^2$ measured by exact message passing and divided by the predicted $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$: the families with finite eighth moments converge to 1.

This is the spine of the paper and it is computation, not fitting. Under the Ornstein–Uhlenbeck channel the Hermite moment of degree m is attenuated exactly by e^{-mt}, so the Fisher information about a feature that first appears at degree m decays as e^{-2mt} and the sequences needed for a fixed precision grow as e^{2mt}. Correlation sits at degree two and shape at degree four, which is why they are lost at different speeds and why the gap between them widens with t. Panel (c) is the sharpest version: the entire cost of approximating a non-Gaussian chain by its covariance twin is $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$, and adding the fourth-order term back closes 0.961 of it at $t = 0.68$ and all of it from $t = 2$. *Compendium, Chapters 3, 5, 10, 13.*

4 What capacity buys

A Gaussian-innovation kernel ($C = 1$) cannot beat the LMMSE of the covariance twin, and does not: it sits on that floor at every sample size. The mixture goes far below it, and the margin is

Table 1: Schedule-pooled relative score error on all sites, mean over 16 seeds (s.e. in the text macros; the isotropic reference is 0.669 at every size). Every arm is scored on identical test bundles. The last column is the same capacity fitted to the innovation law itself—the C-component mixture closest in Kullback–Leibler divergence, an infinite-data reference for the family rather than a bound; † marks fits whose narrowest component is under two grid cells.

	sequences M				
kernel	32	128	512	2048	$M = \infty$
LMMSE floor	0.167	0.167	0.167	0.167	0.167
$C = 1$	0.174	0.168	0.168	0.167	0.167
$C = 2$	0.084	0.050	0.035	0.028	0.039
$C = 4$	0.093	0.053	0.037	0.025	0.0057
$C = 8$	0.099	0.055	0.036	0.023	0.0022†
$C = 16$	0.100	0.056	0.036	0.022	0.0007†

the fourth-order structure and nothing else — across the generalised-Gaussian family the gain is ordered by κ_4 and vanishes exactly at the Gaussian member.

The contrast is not a property of the frozen corner. Repeating it over five correlations and three chain lengths at 16 seeds gives the same answer in all 15 cells, with every interval excluding zero:

Table 2: Capacity across regimes (E13X, 16 seeds). Last column: the paired difference $C = 8$ minus $C = 1$, negative in favour of the mixture. The $C = 1$ column against the LMMSE column is Proposition 1 checked at every cell — they agree to 0.0099 in the worst of the 15.

ρ	L	t_*	$C = 1$	$C = 8$	LMMSE	C=8 − C=1
0.3	16	0.30	0.281	0.051	0.280	−0.230 [-0.234, -0.225]
0.3	32	0.31	0.285	0.042	0.285	−0.242 [-0.246, -0.238]
0.3	64	0.31	0.286	0.033	0.286	−0.253 [-0.257, -0.249]
0.5	16	0.52	0.263	0.060	0.261	−0.202 [-0.209, -0.195]
0.5	32	0.54	0.266	0.045	0.265	−0.221 [-0.225, -0.216]
0.5	64	0.55	0.267	0.037	0.267	−0.230 [-0.235, -0.226]
0.7	16	0.79	0.224	0.050	0.223	−0.174 [-0.180, -0.168]
0.7	32	0.84	0.227	0.043	0.226	−0.184 [-0.191, -0.176]
0.7	64	0.86	0.227	0.029	0.227	−0.198 [-0.202, -0.194]
0.85	16	1.05	0.165	0.046	0.165	−0.120 [-0.125, -0.114]
0.85	32	1.17	0.168	0.036	0.167	−0.131 [-0.135, -0.128]
0.85	64	1.22	0.168	0.029	0.167	−0.139 [-0.141, -0.137]
0.95	16	1.26	0.086	0.039	0.084	−0.047 [-0.053, -0.040]
0.95	32	1.50	0.086	0.026	0.086	−0.060 [-0.063, -0.057]
0.95	64	1.67	0.087	0.025	0.086	−0.061 [-0.065, -0.058]

Two things are worth taking from that table beyond the result. The margin shrinks as ρ rises, in the direction the theory gives: a strongly correlated chain has a nearly Gaussian marginal (excess kurtosis $3(1-\rho^2)/(1+\rho^2)$), so there is less fourth-order structure to buy. And the $C = 1$ arm lands on the LMMSE floor in every one of the 15 cells, which tests the grid, the message passing, the EM and the scoring simultaneously, at fifteen independent configurations. The sixteen-seed run also reproduces its eight-seed predecessor exactly: 480 shared rows recomputed in a different directory, from a different deployment, on a different node, agree to 0.

One correction belongs with this result rather than in a footnote. The capacity sweep in the archival record — the one Chapter 10 of the thesis rests on, $C \in \{1, 2, 4, 8, 16\}$ at $M \in \{128, 512\}$ over sixteen seeds — is void, for a reason prior to any reading of it: its training draw re-created the random generator for every sequence, so every arm at every capacity and seed was fitted to a single chain of L sites, its validation and test bundles alone drawn correctly. Its own numbers carry

the signature: a correctly fitted $C = 1$ kernel must sit on the LMMSE floor 0.168, which every cell of the table above reaches to 0.0099, and the old sweep's single-Gaussian arm scatters across a range several times wider. The earlier diagnosis — that a stopping rule explained the conclusion — is withdrawn with it; the asymmetry is real but is EM's behaviour on one chain. Nothing in this section depends on that sweep: the runs behind it draw every chain independently and assert that they did. *Compendium, Chapters 13, 15, 17.*

5 Structure against networks, at equal information

This is the claim a referee will press hardest, so it is worth being precise about what is being compared. Both sides see exactly one noised view of each sequence, drawn once. Neither side ever sees the clean chain. Neither side uses the true score to choose when to stop: the networks select on the implicit objective of a held-out noisy bundle, EM–BP on the held-out one-view evidence.

Table 3: Interface C across correlation and length (E14, 16 seeds, $M = 2048$, networks re-screened inside each regime). Columns: the LMMSE oracle; EM–BP with its iteration count chosen by held-out evidence (ev.) and by validation against the true kernel (bp); the best network trained by implicit score matching on the same one-view bundle (C); the best network given the clean sequence with one fixed corruption (B); and the paired factor of the C network over the evidence-selected EM–BP, with a percentile bootstrap over seeds.

ρ	L	LMMSE	EM–BP (ev.)	EM–BP (bp)	best net (C)	best net (B)	factor C / EM–BP (ev.)
0.5	32	0.265	0.026	0.024	0.137	0.129	5.7 [4.9, 6.7]
0.5	128	0.268	0.020	0.018	0.089	0.079	4.6 [4.1, 5.0]
0.85	32	0.167	0.024	0.021	0.170	0.156	8.3 [6.6, 10.3]
0.85	128	0.168	0.015	0.015	0.106	0.095	8.0 [6.7, 9.2]
0.95	32	0.086	0.021	0.019	0.196	0.128	10.3 [8.8, 11.9][†6]
0.95	128	0.087	0.017	0.016	0.105	0.094	6.4 [6.0, 6.9]

EM–BP is ahead in every one of the 6 cells and in every one of the 16 seeds of each, by factors from 4.6 to 10.3. The factor is also conservative: the evidence-selected EM–BP is still improving when its iteration ladder runs out in 56 of 96 of the cells, so its error there is an upper bound on what a longer ladder would give.

Three readings of the table matter more than the headline number.

- **The advantage is smallest where the chain carries the most higher-order structure.** At $\rho = 0.5$ the marginal's excess kurtosis is 1.80 and the factor is 5.7; at $\rho = 0.95$ the kurtosis is 0.15 and the factor is larger. Above $\rho = 0.85$ the two chain lengths part company, so we report the direction and not a law.
- **Length does not rescue the network.** At $L = 128$ each noised view carries four times as many sites, and both arms improve — the network from 0.170 to 0.106 at $\rho = 0.85$, EM–BP likewise — so the factor survives rather than being an artefact of short chains.
- **Where the network sits relative to the second-order floor is a statement about the floor.** The network's error changes little across the sweep while the floor falls steeply with ρ. So the network is below the floor at (0.5, 32), (0.5, 128), (0.85, 128) and above it at (0.85, 32), (0.95, 32), (0.95, 128) — at $\rho = 0.95$, $L = 32$ by $+0.110$ $[+0.108, +0.112]$, against an exact radius-8 floor of 0.025 at that correlation. Where the covariance carries most of the score energy, one view per sequence is not enough for a network to learn even the covariance, while the estimator that knows the factorisation recovers it.

And it is not about heavy tails. Repeated at the frozen point with a *platykurtic* innovation — a symmetric two-component mixture, kurtosis on the other side of Gaussian — the factor is 8.8 $[7.4, 10.2]$ over 16 seeds, with the best network (transformer) at 0.159 against a floor of 0.082 — 1.93

times above it. What explicit structure buys is fourth-order structure of either sign. *Compendium, Chapters 8, 17, 19.*

6 Where the advantage ends

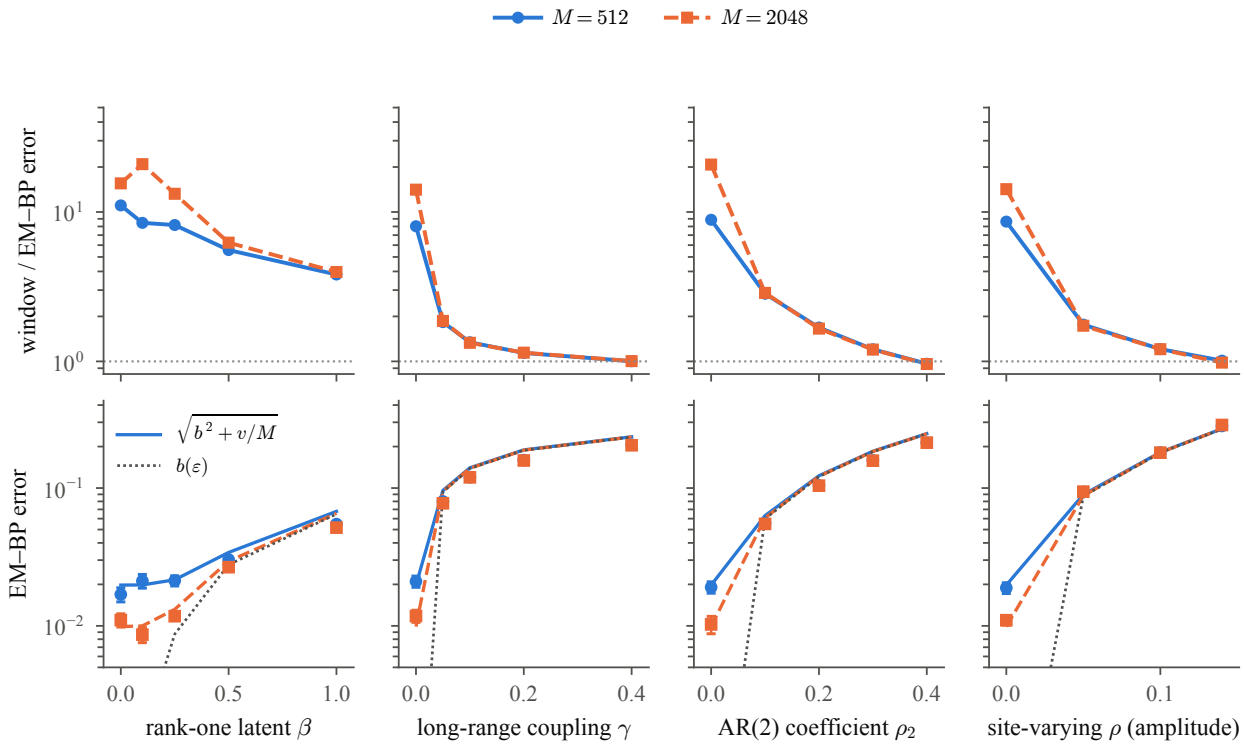


Figure 2: *Top row:* ratio of a screened radius-2 window head's pooled error to EM–BP's against the strength of four departures from the first-order Markov assumption, at two sample sizes; above the dotted line the misspecified chain estimator still wins. *Bottom row:* EM–BP's own error at both sizes (points, paired standard errors) against the closed-form bias $b(\varepsilon)$ of the Kullback–Leibler projection of the perturbed truth onto the chain family (dotted) and the finite-sample prediction $\sqrt{b(\varepsilon)^2 + v/M}$ (lines), with v read off the well-specified cells. The bias is predicted in closed form, not fitted to these points.

The estimator assumes a first-order Markov chain, and the honest question is how wrong that may be before the advantage goes. Each departure is Gaussian, so its score is exact and every arm is scored against the truth of the perturbed model rather than against the chain. Adding an order (AR(2)) or making the chain inhomogeneous (a site-dependent ρ) reverses the ordering inside the swept range, at 0.3–0.4 and 0.1–0.14 respectively. A rank-one global latent and a long-range precision coupling never reverse it: the first is absorbed by the fitted ρ, the second decays to a tie at the widest coupling swept. The two sample sizes separate the two failure modes exactly as the theory says they should — at zero strength the error halves between $M = 512$ and 2048, which is estimation; past the break-even it does not move at all, which is bias, and no amount of data will move it. *Compendium, Chapters 18, 19.*

7 Does score accuracy become sample quality

Driven through the same reverse diffusion on identical Brownian paths, so that the only difference between the arms is the score, the fitted kernel produces sequences with innovation excess kurtosis 1.680 against the reference sample's 1.94 — close to what the *exact* score itself achieves under the same sampler (1.863), so most of the residue is the discretisation of the reverse process and not

Table 4: Generated samples against the true law at $t_{\min}$: mean log-likelihood gap $\Delta\ell$ to the reference sample under the true $P_{t_{\min}}$ (exact evidence; the exact arm's gap is the sampler's discretisation floor), innovation excess kurtosis $\hat{\kappa}_4$, lag fourth cumulant, sliced W_2; the row marked *reference* carries the reference sample's own statistics. Mean over seeds.

M	arm	$\Delta\ell$	$\hat{\kappa}_4$	lag-κ_4	W_2
128	*reference*	–	1.94	0.000	–
	exact	0.074	1.86	0.000	0.051
	em–bp	0.204	1.68	0.000	0.078
	lmmse	1.761	0.00	0.000	0.053
	isotropic	46.241	0.00	0.006	0.244
	net:bigru	0.752	1.10	0.006	0.160
	net:window	1.284	1.27	0.004	0.154
2048	*reference*	–	1.94	0.000	–
	exact	0.070	1.87	0.000	0.051
	em–bp	0.164	1.80	0.001	0.055
	lmmse	1.766	0.00	0.000	0.053
	isotropic	46.295	0.00	0.010	0.244
	net:bigru	0.385	1.55	0.003	0.167
	net:window	1.115	1.46	0.003	0.139

the estimate. Its log-likelihood gap under the true law is 0.204 against the exact arm's 0.074. The covariance-matched score gives -0.001 — an essentially Gaussian chain, the fourth-order structure gone altogether — at a gap of 1.761, and the isotropic score 46.241. Score error and sample quality move together, which is what licenses reading the rest of the paper as being about generative models and not only about an estimation problem. *Compendium, Chapters 9, 21.*

8 Where the evidence stands

Every number above is a macro read from a run whose output directory is pinned by SHA-256 in its registry entry, together with the commit that produced it and the seeds it used; the manuscript refuses to build if a generated number disagrees with the artefact it came from, and a second, independent recomputation of every headline number runs beside it. The confirmatory experiments are frozen: the information decay and its constants, the capacity sweep and its infinite-data reference, the six-architecture comparison with its screen, generation, misspecification, the quadrature envelope, the regime sweeps in both capacity and the matched comparison, and the platykurtic cell. One run is still finishing on the cluster — the sample-complexity ladder, whose last two cells of four hundred and eighty are recomputing — and one paragraph waits on it.